

Automatic Pet Feeder

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Abstract: This article presents a modern solution for remote pet feeding [5]. Some automatic feeders use gravity to dispense food from a chamber as it is eaten from a dish below. Most of the pet feeders jam during the process. It was also observed that the programmable pet feeders over fed the pets [2]. They were inconsistent in how much over the amount of food. The work done focuses on dispensing an accurate amount of food [6] as required by the user as well as maximizing user interaction with the help of motion detectors and other such sensors. Microcontrollers integrated with Wi-fi modules are utilized to provide better user interaction and real-time monitoring [4]. After testing and troubleshooting the final working model was developed to work efficiently according to user requirements. Automated embedded system integration [1] was explored along with working of sensors and IoT cloud connection.

Keywords Automatic pet feeders, Automated embedded system integration, IoT cloud connection, Microcontrollers.

INTRODUCTION

In the modern era, where the majority of the people work 9 to 5 jobs, a pet feeder has become a common need for those who leave their pets at home on a day-to-day basis. There have been many designs proposed previously, whether it is timer based, sensor based or even manual. The use of sensors is a huge added benefit when it comes to feed systems because it automates the process completely, allowing for minimal human interference. Oftentimes, a source of stress for these animals can be not getting fed the proper amounts of feed or not getting fed on time. Inconsistent diet is not good for a pet. In addition, many times feed will come in an airtight bag that pet owners can forget to close completely after feeding pets. This leads to things like growing mold spores and bacteria cultures in the feed itself.

Feed systems for pets can be designed as an automated device [7], that allow the user to predetermine when the feed is distributed from a pre-set list of parameters that may include per time basis, weight basis, or proximity basis which takes position of pet in account with respect to the feeding system device. We added various sensors and enabled cloud connection to automate the entire process. In addition to having a single sensor in a system like this, it is possible to incorporate other aspects to the system, to make it more efficient and user friendly.

Thus, we came up with our implementation of this idea that covers as many needs as possible. An automated system that can work independently as well as connect to the internet to give you remote access to your device [3]. An implementation that allows the user to control the amount of food in each server and the ability to set up a timer. We also added other ways of feeding your pet and to make sure your pet receives the food.

PROPOSED SYSTEM DESIGN

HARDWARE

- Node-MCU- Acts as main controller, integrates the entire circuit, as well as enables Wi-Fi connection
- PIR sensor- Used to check presence of pet
- Buzzer- Used in order to signal food dispensing
- Ultrasonic sensor- Used to measure the height of food in the measurement unit in order to determine the amount of food
- Servo motors- To control the food dispensing process

SOFTWARE

- Arduino IoT Cloud- Open-source platform
Main platform that was used to upload the code to the Node-MCU, connect the system to the internet and integrate the dashboard to the total setup.

FLOW CHART

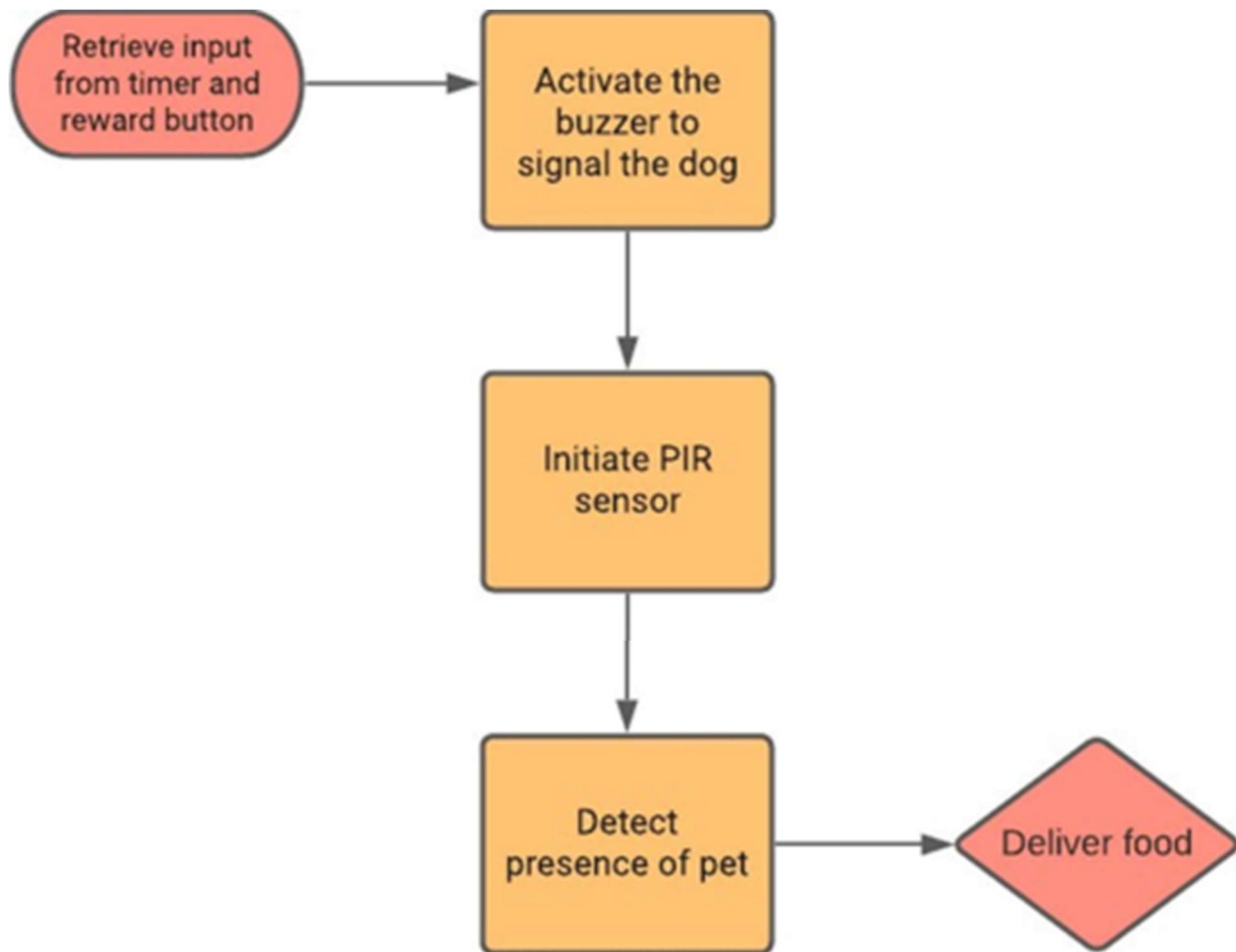


FIGURE.1. This flowchart explains how information is transferred and decisions are made.

CIRCUIT SCHEMATIC

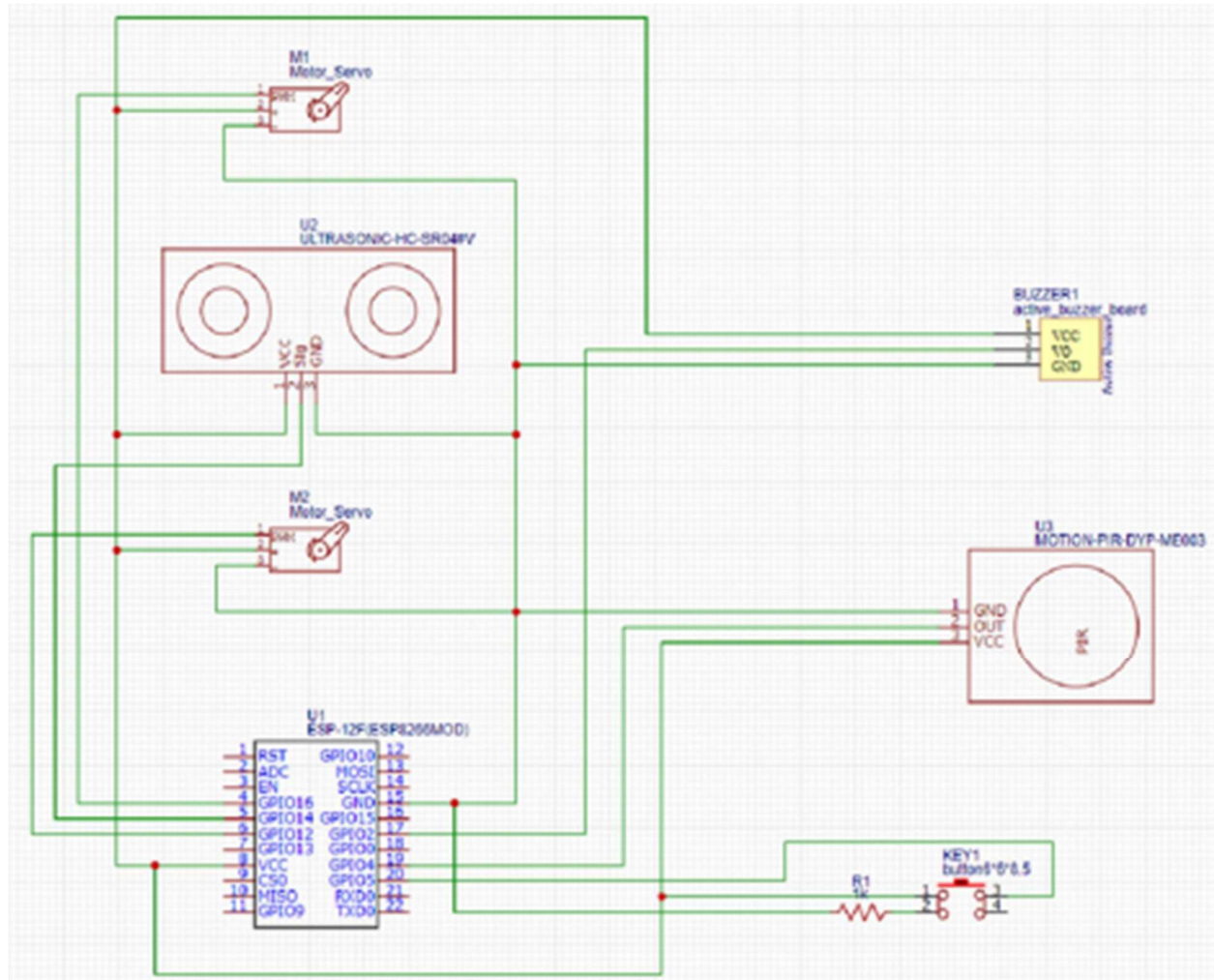


FIGURE.2. An electrical circuit is represented graphically in this circuit diagram.

DESIGN

Designing of the product is done in such a manner that is easy to maintain and is user friendly as well.

FRONT VIEW

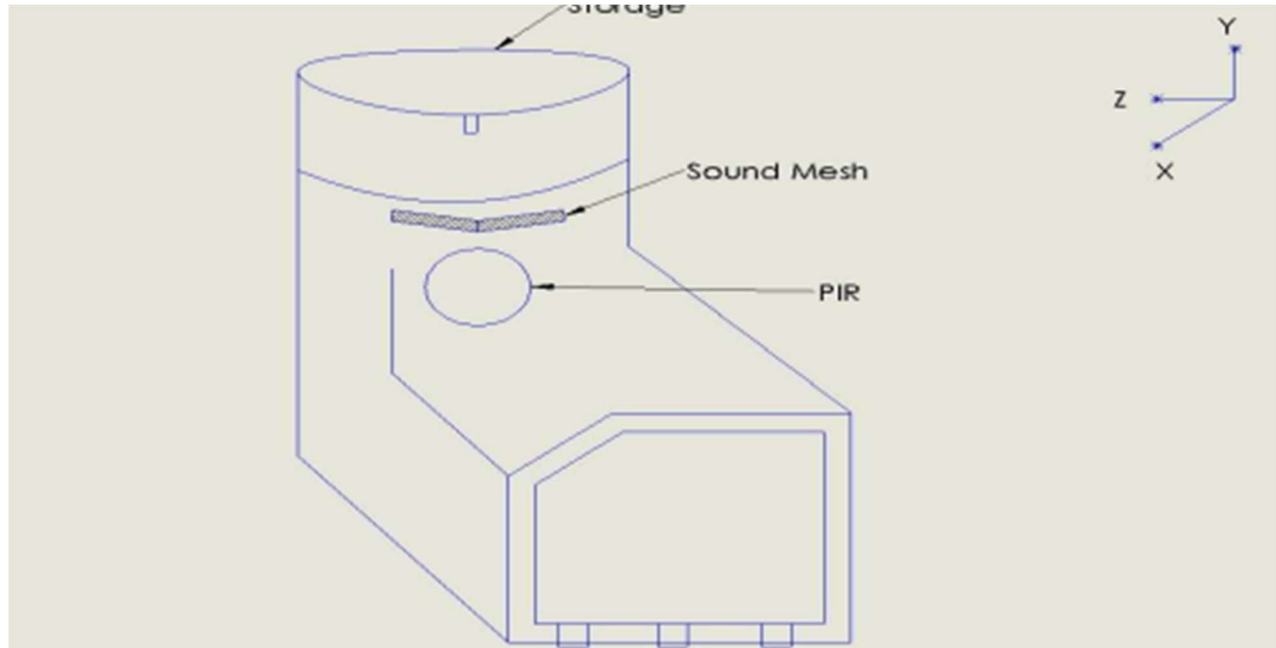


FIGURE.3. Front view of the hardware implementation is shown above

SIDE VIEW

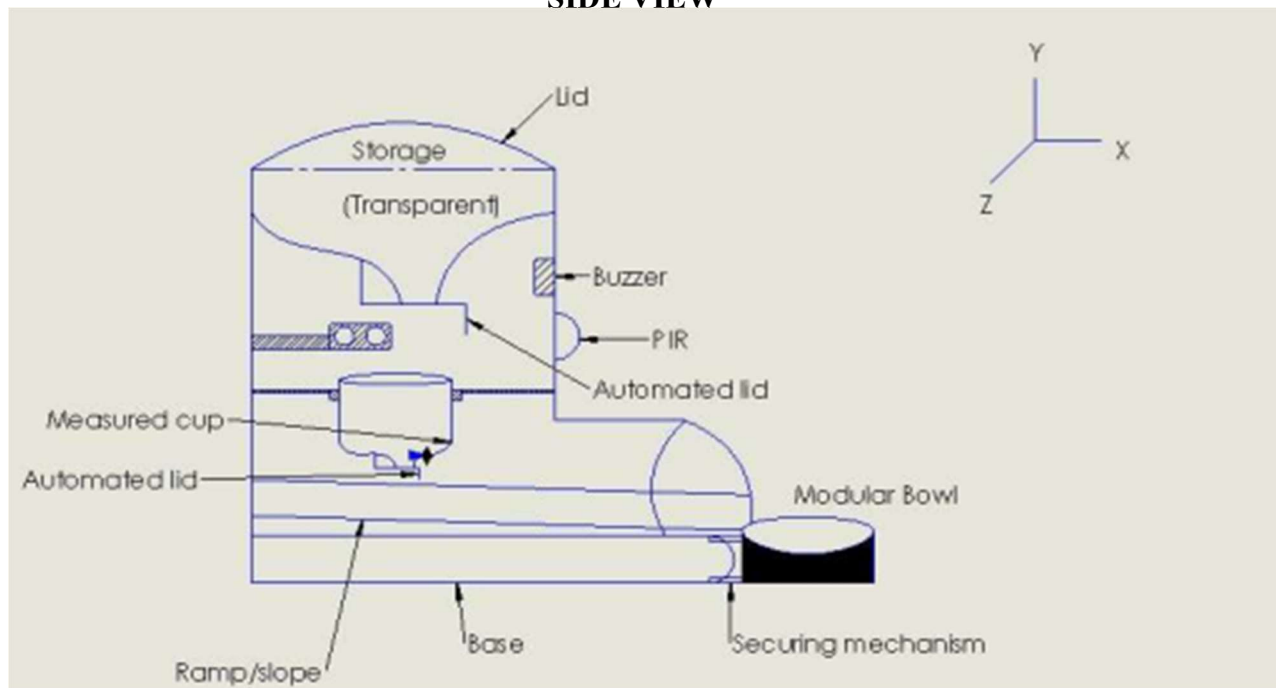


FIGURE.4. Side view of the hardware implementation is shown above

RESULTS

DASHBOARD

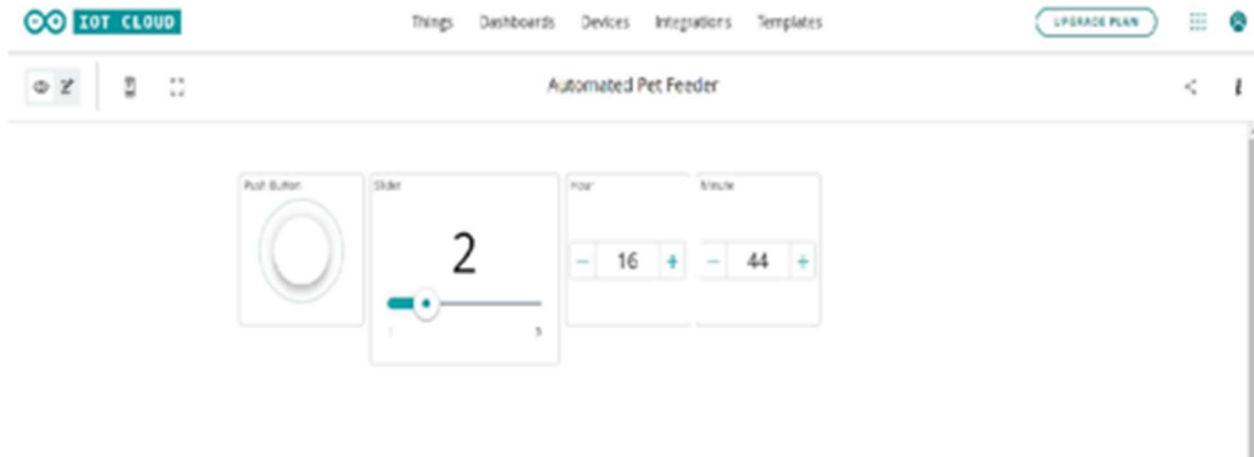


Fig.5(a). IOT Cloud dashboard

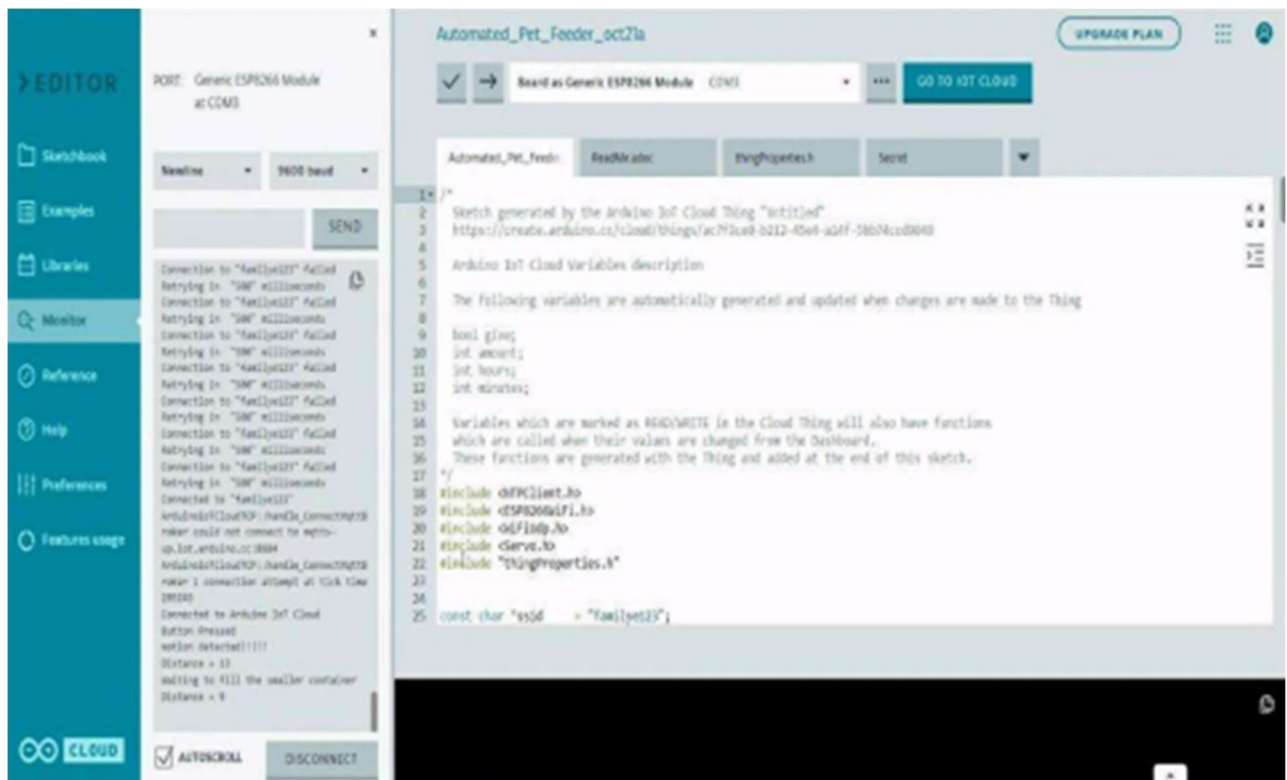


Fig.5(b). IOT Cloud dashboard

CONCLUSION AND FUTURE WORK

The system was implemented using the acquired hardware components. The code was tested and modified according to requirements. Automated pet feeder system offers an effective way to monitor everyday pet feeding schedules. This system can greatly benefit pet owners and pets equally. Since the system is user operated, the feeder can be switched on or off whenever needed by the owner and the reward button integrated into the IoT cloud can enable the user to authorize treats whenever they deem it fit. It is demonstrated that it is possible to create a system that is self-sufficient and works well with the user interface to serve the purpose of scheduled pet feeding. The proposed system provides the means for real time monitoring and efficient feeding even in the absence of the owner. The results obtained present the key improvement of the pet care system involved with the IoT technology, but also meet the needs of pet owners.

FUTURE WORK

- A manual reward button can be integrated into the system that can be used as a way to encourage positive behaviour in pets by the respective owners.
- The IoT communication can be enhanced, to facilitate added options to the user, enabling them to define timings and other variables that are integrated in the system.
- Additional modules can be added without affecting the pre-existing setup. This allows the flexibility and easy maintenance of the developed system.
- Different coding languages and algorithms can be implemented to better accommodate versatility of code and enable easier modification according to requirement.
- The overall design of the system can be improved to accommodate a more user-friendly interface.

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